✪ Psychological Statistics ✪

Multi-Group: Nonparametric & Robust Methods

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In [ ]:
```
### [ Setup the working directory ]

setwd("/Users/wesley/[Course]/Python/R_Script")
getwd()
```

In [2]:
```
### [ Loading the required libraries ]

library(ggpubr)
library(pastecs)
library(reshape2)
library(car)
library(rstatix)
library(WRS2) # https://rdrr.io/cran/WRS2/
```

---

(1) Kruskal-Wallis Tests (One-way Independent ANOVA)

Example: Bushtucker

[Hypothesis] Rat growth can be predicted by dietary tannin concentration. *(2-tailed)*

- Null hypothesis H_0 **:** No linear relationship between rat growth and tannin concentration $\rightarrow \beta = 0$
- Alternative hyp. H_1 **:** There is (linear) relationship between rat growth and tannin concentration $\rightarrow \beta \neq 0$

In [3]:
```
### [ Step.1 ] Load data

data("PlantGrowth")
set.seed(1234)
PlantGrowth %>% sample_n_by(group, size = 3)

levels(PlantGrowth$group)

# If the levels are not automatically in the correct order, re-order them as fo
PlantGrowth <- PlantGrowth %>%
  reorder_levels(group, order = c("ctrl", "trt1", "trt2"))

#----- Descriptive stat. & plots -----
PlantGrowth %>%
  group_by(group) %>%
  get_summary_stats(weight, type = "common")
```

A tibble: 9 × 2

weight	group
<dbl>	<fct>
5.14	ctrl
4.61	ctrl
4.50	ctrl
4.32	trt1
5.87	trt1
3.83	trt1
5.50	trt2
5.12	trt2
4.92	trt2

'ctrl' · 'trt1' · 'trt2'

A tibble: 3 × 11

group	variable	n	min	max	median	iqr	mean	sd	se	ci
<fct>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
ctrl	weight	10	4.17	6.11	5.155	0.743	5.032	0.583	0.184	0.417
trt1	weight	10	3.59	6.03	4.550	0.662	4.661	0.794	0.251	0.568
trt2	weight	10	4.92	6.31	5.435	0.467	5.526	0.443	0.140	0.317

In [2]:
```
### [ Step.2 ] Check assumptions
```

In [4]:
```
### [ Step.3-1 ] K-W test

res.kruskal <- PlantGrowth %>% kruskal_test(weight ~ group)
res.kruskal
```

A rstatix_test: 1 × 6

	.y.	n	statistic	df	p	method
	<chr>	<int>	<dbl>	<int>	<dbl>	<chr>
1	weight	30	7.988229	2	0.0184	Kruskal-Wallis

In [5]:
```
### [ Step.3-2 ] Post-hoc Tests

#----- pairwise comparisons -----

# (a) Wilcoxon's Test
pwc <- PlantGrowth %>%
  wilcox_test(weight ~ group, p.adjust.method = "bonferroni")
pwc

# (b) Dunn's Test
pwc2 <- PlantGrowth %>%
  dunn_test(weight ~ group, p.adjust.method = "bonferroni")
pwc2
```

A rstatix_test: 3 × 9

	.y.	group1	group2	n1	n2	statistic	p	p.adj	p.adj.signif
	<chr>	<chr>	<chr>	<int>	<int>	<dbl>	<dbl>	<dbl>	<chr>
1	weight	ctrl	trt1	10	10	67.5	0.199	0.597	ns
2	weight	ctrl	trt2	10	10	25.0	0.063	0.189	ns
3	weight	trt1	trt2	10	10	16.0	0.009	0.027	*

A rstatix_test: 3 × 9

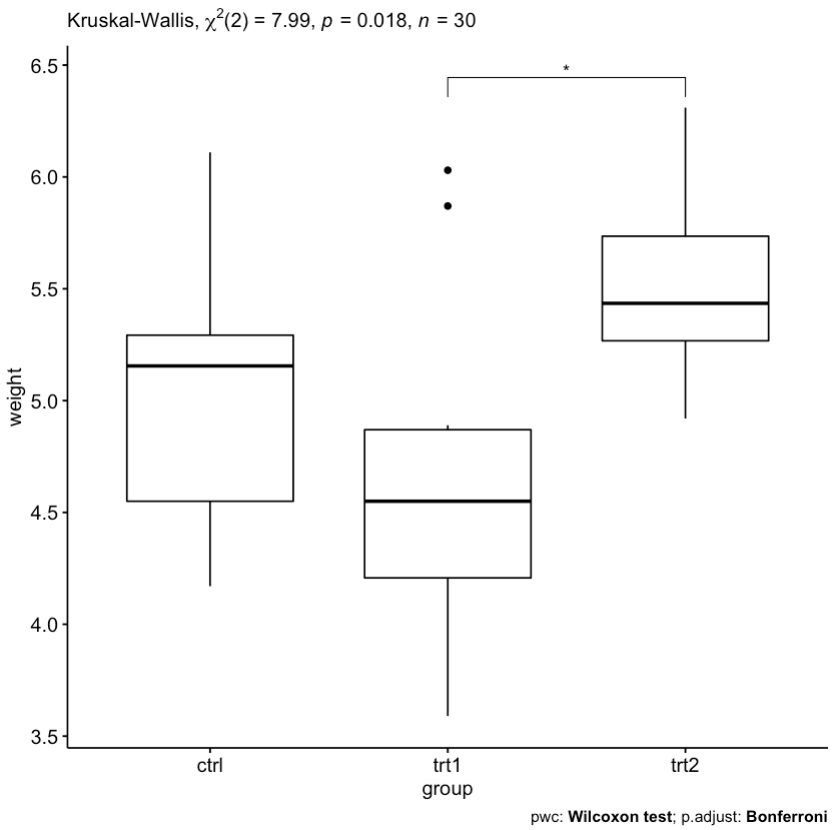
	.y.	group1	group2	n1	n2	statistic	p	p.adj	p.adj.signif
	<chr>	<chr>	<chr>	<int>	<int>	<dbl>	<dbl>	<dbl>	<chr>
1	weight	ctrl	trt1	10	10	-1.117725	0.26368427	0.79105280	ns
2	weight	ctrl	trt2	10	10	1.689290	0.09116394	0.27349183	ns
3	weight	trt1	trt2	10	10	2.807015	0.00500029	0.01500087	*

In [6]:
```
### [ Step.4 ] Effect size

PlantGrowth %>% kruskal_effsize(weight ~ group)
```

A rstatix_test: 1 × 5

	.y.	n	effsize	method	magnitude
	<chr>	<int>	<dbl>	<chr>	<ord>
1	weight	30	0.2217862	eta2[H]	large

In [7]:
```
### [ Step.5 ] Visualization

pwc <- pwc %>% add_xy_position(x = "group")
ggboxplot(PlantGrowth, x = "group", y = "weight") +
  stat_pvalue_manual(pwc, hide.ns = TRUE) +
  labs(subtitle = get_test_label(res.kruskal, detailed = TRUE),
    caption = get_pwc_label(pwc))
```



~ *Report* ~

- There was a significant differences between treatment groups as assessed using the Kruskal-Wallis test (p = 0.018).
- Pairwise Wilcoxon test between groups showed that only the difference between trt1 and trt2 group was significant (Wilcoxon's test, p = 0.027)

---

(2) Friedman Test (One-way repeated-measure ANOVA)

Example: Self esteem

[Hypothesis] We repeatedly tracing the self-esteem from child, adolescent and adult, where the three time points were marked as t1, t2 and t3. *(2-tailed)*

- Null hypothesis H_0 **:** No prominent changes of self-esteem along the 3 time points.
- Alternative hyp. H_1 **:** There self-esteem changes along the 3 time points.

In [8]:
```
### [ Step.1 ] Data Loading

data("selfesteem", package = "datarium")
head(selfesteem, 3)

names(selfesteem)

# Convert id and time into factor variables (Long form)
selfesteem <- selfesteem %>%
  gather(key = "time", value = "score", t1, t2, t3) %>%
  convert_as_factor(id, time)
head(selfesteem, 3)

#----- Descriptive stat. & plots -----
selfesteem %>%
  group_by(time) %>%
  get_summary_stats(score, type = "mean_sd")
```

A tibble: 3 × 4

id	t1	t2	t3
<int>	<dbl>	<dbl>	<dbl>
1	4.005027	5.182286	7.107831
2	2.558124	6.912915	6.308434
3	3.244241	4.443434	9.778410

'id' · 't1' · 't2' · 't3'

A tibble: 3 × 3

id	time	score
<fct>	<fct>	<dbl>
1	t1	4.005027
2	t1	2.558124
3	t1	3.244241

A tibble: 3 × 5

time	variable	n	mean	sd
<fct>	<chr>	<dbl>	<dbl>	<dbl>
t1	score	10	3.140	0.552
t2	score	10	4.934	0.863
t3	score	10	7.636	1.143

In [7]:
```
### [ Step.2 ] Check assumptions
```

In [9]:
```
### [ Step.3-1 ] Friedman test

res.fried <- selfesteem %>% friedman_test(score ~ time |id)
res.fried
```

A rstatix_test: 1 × 6

	.y.	n	statistic	df	p	method
	<chr>	<int>	<dbl>	<dbl>	<dbl>	<chr>
1	score	10	18.2	2	0.0001116658	Friedman test

In [10]:
```
### [ Step.3-2 ] post-hoc test

#----- pairwise comparisons -----

# (a) Wilcoxon's Test
pwc <- selfesteem %>%
  wilcox_test(score ~ time, paired = TRUE, p.adjust.method = "bonferroni")
pwc

# (b) Sign Test
pwc2 <- selfesteem %>%
  sign_test(score ~ time, p.adjust.method = "bonferroni")
pwc2
```

A rstatix_test: 3 × 9

	.y.	group1	group2	n1	n2	statistic	p	p.adj	p.adj.signif
	<chr>	<chr>	<chr>	<int>	<int>	<dbl>	<dbl>	<dbl>	<chr>
1	score	t1	t2	10	10	0	0.002	0.006	**
2	score	t1	t3	10	10	0	0.002	0.006	**
3	score	t2	t3	10	10	1	0.004	0.012	*

A rstatix_test: 3 × 10

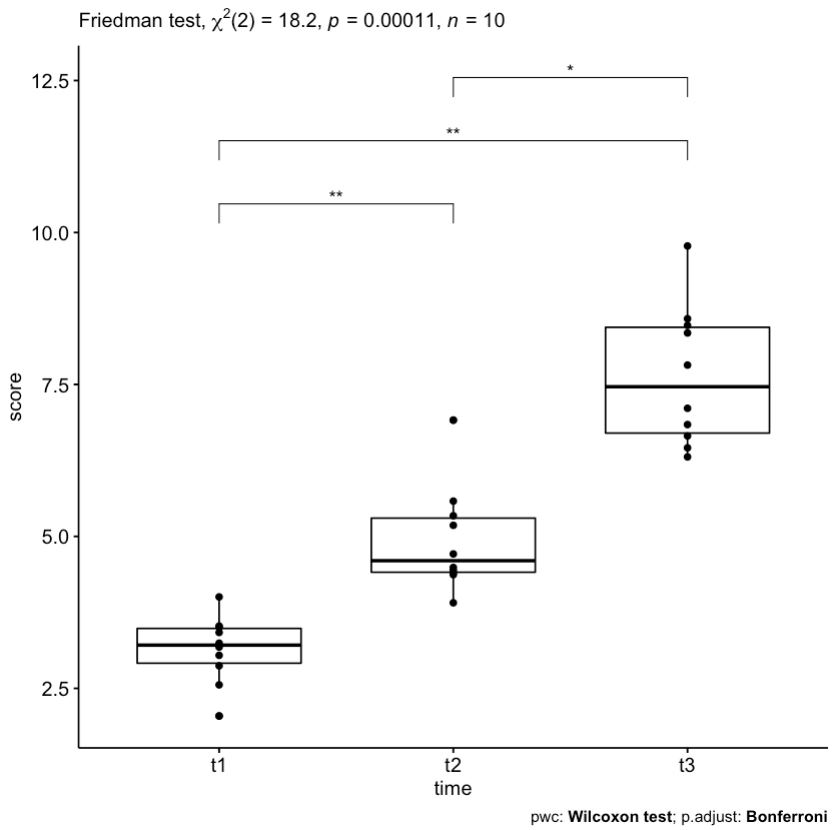
	.y.	group1	group2	n1	n2	statistic	df	p	p.adj	p.adj.signif
	<chr>	<chr>	<chr>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	score	t1	t2	10	10	0	10	0.002	0.006	**
2	score	t1	t3	10	10	0	10	0.002	0.006	**
3	score	t2	t3	10	10	1	10	0.022	0.064	ns

In [11]:
```
### [ Step.4 ] Effect size

selfesteem %>% friedman_effsize(score ~ time |id)
```

A rstatix_test: 1 × 5

	.y.	n	effsize	method	magnitude
	<chr>	<int>	<dbl>	<chr>	<ord>
1	score	10	0.91	Kendall W	large

In [12]:
```
### [ Step.5 ] Visualization

pwc <- pwc %>% add_xy_position(x = "time")
ggboxplot(selfesteem, x = "time", y = "score", add = "point") +
  stat_pvalue_manual(pwc, hide.ns = TRUE) +
  labs(subtitle = get_test_label(res.fried,  detailed = TRUE),
    caption = get_pwc_label(pwc))
```



~ *Report* ~

- The self-esteem score was significantly different at the different time points using Friedman test, chi-squared(2) = 18.2, p = 0.00011.
- Pairwise Wilcoxon signed rank test between groups revealed significant differences in self esteem between t1 and t2 (p = 0.006); t1 and t3 (0.006); t2 and t3 (0.012).

---

(3) Bootstrapping of T-test

Example B - Repeated-measure ANOVA

[Hypothesis] Album sales can be affected by 3 factors: advert, airplay and attractiveness. *(2-tailed)*

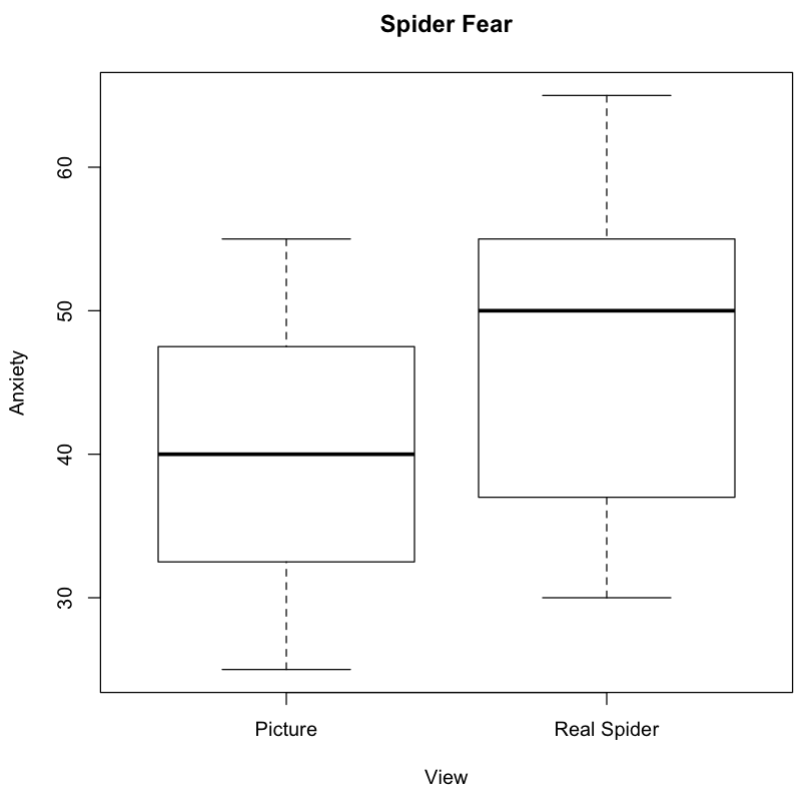
- Null hypothesis H_0 **:** No prominent relations can be observed among album sales and the 3 factors (all βs = 0).
- Alternative hyp. H_1 **:** The album sales can be predicted by at least one of the 3 factors (βs ≠ 0).

In [13]:
```
### [ Step.1 ] Data Loading

# → Loading the dataset
  spiderLong<-read.delim("SpiderLong.dat")

  # Descriptive statistics
  names(spiderLong)
  levels(spiderLong$Group)
  table(spiderLong$Group)
  boxplot(Anxiety~Group, ylab="Anxiety", data=spiderLong,
        xlab="View", main="Spider Fear")
```

'Group' · 'Anxiety'

'Picture' · 'Real Spider'

```
   Picture Real Spider
        12          12
```



In [ ]:
```
### [ Step.2 ] Check assumptions

#----- None -----#
```

In [14]:
```
# [ Step.3 ] Bootstrapping (T-test)

# A. calculate the difference in sample means
diff_mean <- abs(mean(spiderLong$Anxiety[spiderLong$Group=="Picture"])-mean(spid
with(spiderLong, tapply(Anxiety, Group, mean))
diff(with(spiderLong, tapply(Anxiety, Group, mean)))
#diff_mean <- abs(diff(with(spiderLong, tapply(Anxiety, Group, mean))))
```

Picture: 40 **Real Spider:** 47

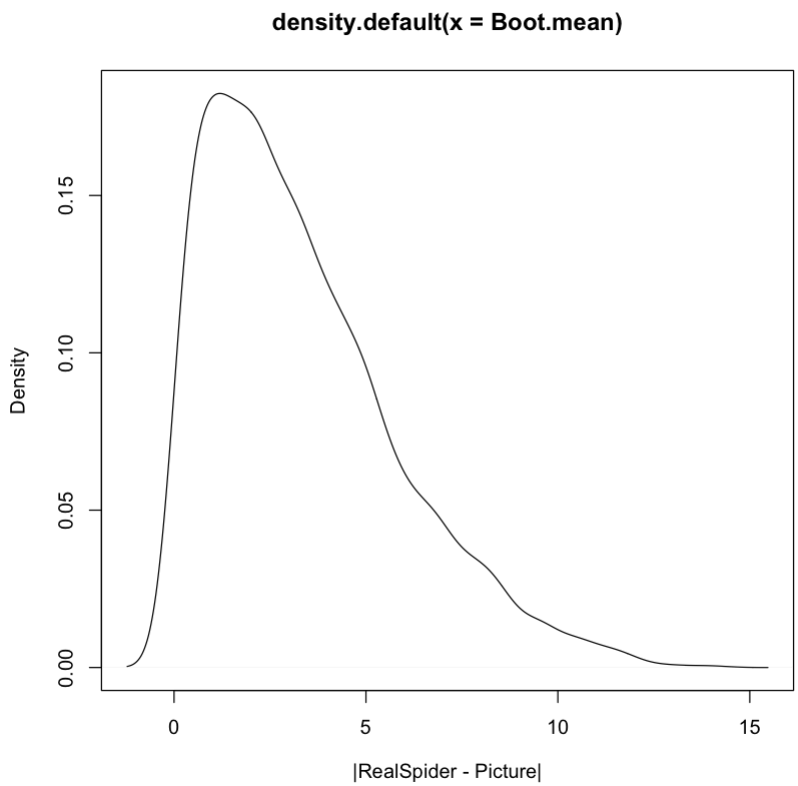
Real Spider: 7

In [15]:
```
### [ Step.3 ] Bootstrapping (T-test)

# B. set parameters for bootstrapping
  set.seed(123)  # for reproducibility
  num <- length(spiderLong$Anxiety) # number of observations in each sample
  Nb = 5000 # re-sample number for bootstrapping
  BootSamples <- matrix(sample(spiderLong$Anxiety, size=num*Nb, replace=TRUE),
                        nrow=num, ncol=Nb)
  dim(BootSamples)
```

24 · 5000

In [16]:
```
### [ Step.3 ] Bootstrapping (T-test)

# C. Perform Bootstrapping method
  Boot.mean <- rep(0,Nb) #initialize the bootstrap vector for TEST-STATS

  for (i in 1:Nb){
    Boot.mean[i] <- abs(mean(BootSamples[1:12,i])-mean(BootSamples[13:24,i]))
  }
```

In [17]:
```
### [ Step.3 ] Bootstrapping (T-test)

# D. Calculate p-value and C.I. via probability distribution (histogram)
  p_mean = mean(Boot.mean>=diff_mean)
  CI_1=quantile(Boot.mean, prob=.025)
  CI_2=quantile(Boot.mean, prob=.975)
  plot(density(Boot.mean), xlab=expression("|RealSpider - Picture|"))
```



In [18]:
```
### < 2.Robust method of T-test (WRS2) >

  library(WRS2)

  #---{Independent-sample T-test}---#
  set.seed(123)
  yuenbt(Anxiety~Group, data = spiderLong, tr=0.1, nboot=2000)
  # effect size
  yuen.effect.ci(Anxiety~Group, data = spiderLong, tr = 0.1, nboot = 2000, alph
```

```
Call:
yuenbt(formula = Anxiety ~ Group, data = spiderLong, tr = 0.1,
    nboot = 2000)

Test statistic: -1.5213 (df = NA), p-value = 0.1275

Trimmed mean difference:  -6.9
95 percent confidence interval:
-15.9233     2.1233
```

$effsize

0.436763664919261

$alpha

0.05

$CI

0 · 0.841499662385392

In [19]:
```
### < 2.Robust method of T-test (WRS2) >

#---{Dependent-sample T-test}---# (note: different type in input structure)
  spiderLong$subj<- c(rep(seq(1,12),2))
  spiderWide <- reshape2::dcast(spiderLong, variable~Group, value.var = "Anxiet
  set.seed(123)
  yuend(spiderWide$Picture, spiderWide$`Real Spider`)
  # effect size
  dep.effect(spiderWide$Picture, spiderWide$`Real Spider`)
```

...

~ *Report* ~

- **The total album sales are positively associated with the advertising budget ($\beta_{Standardized}$ = 0.51, p < .001), the plays on radio ($\beta_{Standardized}$ = 0.51, p < .001), and the attractiveness ($\beta_{Standardized}$ = 0.19, p < .001). This multiple regression model explains 66% of total variance in album sales.**

---

(4) Robust methods of 1-way ANOVA

Example B:

[Hypothesis] **Album sales can be affected by 3 factors: advert, airplay and attractiveness.** ***(2-tailed)***

- Null hypothesis ***H_0 :*** No prominent relations can be observed among album sales and the 3 factors (all βs = 0).
- Alternative hyp. ***H_1 :*** The album sales can be predicted by at least one of the 3 factors (βs ≠ 0).

In [20]:
```r
#-----< 1-way ANOVA >-----#

  libido<-c(3,2,1,1,4,5,2,4,2,3,7,4,5,3,6)
  dose<-gl(3,5, labels = c("Placebo", "Low Dose", "High Dose"))
  viagraData<-data.frame(dose, libido)

  #-> 1-way ANOVA
  t1way(libido ~ dose, data = viagra) # F stat, DOF1, DOF2, p-value
  lincon(libido ~ dose, data = viagra) #-> post hoc

  #-> 1-way ANOVA with bootstrap
  t1waybt(libido ~ dose, tr = .2, nboot = 2000) # F stat & p-value
  mcppb20(libido ~ dose, tr = .2, nboot = 2000)
```

```
Call:
t1way(formula = libido ~ dose, data = viagra)

Test statistic: F = 3
Degrees of freedom 1: 2
Degrees of freedom 2: 4
p-value: 0.16

Explanatory measure of effect size: 0.79
Bootstrap CI: [0.3; 1.43]

Call:
lincon(formula = libido ~ dose, data = viagra)

                 psihat ci.lower ci.upper p.value
placebo vs. low      -1 -5.31858  3.31858 0.43533
placebo vs. high     -3 -7.31858  1.31858 0.18051
low vs. high         -2 -6.31858  2.31858 0.31660

Warning message in t1waybt(libido ~ dose, tr = 0.2, nboot = 2000):
“Some bootstrap estimates of the test statistic could not be computed.”

Call:
t1waybt(formula = libido ~ dose, tr = 0.2, nboot = 2000)

Effective number of bootstrap samples was 1273.

Test statistic: 3
p-value: 0.08877
Variance explained: 0.623
Effect size: 0.789

Call:
mcppb20(formula = libido ~ dose, tr = 0.2, nboot = 2000)

                       psihat ci.lower ci.upper p-value
Placebo vs. Low Dose       -1 -3.33333  1.33333   0.293
Placebo vs. High Dose      -3 -5.33333 -0.33333   0.003
Low Dose vs. High Dose     -2 -4.33333  0.66667   0.082
```

In [21]:
```
#-----< 1-way repeated-measure ANOVA >-----#

  bushData<-read.delim("Bushtucker.dat", header = TRUE)
  bushLong<-reshape2::melt(bushData, id.var = "participant", variable.name = "F
                           value.name = "Time")

  #Trimmed means
  rmanova(bushLong$Time, bushLong$Food, bushLong$participant)
  rmmcp(bushLong$Time, bushLong$Food, bushLong$participant) # post-hoc pairwise

  #bootstrap
  attach(bushLong)
  rmanovab(Time, Food, participant, nboot = 2000)
  pairdepb(Time, Food, participant, nboot = 2000) # post-hoc pairwise
```

...

In [4]:
```
#-----< 1-way ANCOVA >-----#

  invisibilityData<-read.delim("CloakofInvisibility.dat", header = TRUE)
  invisibilityData$cloak<-factor(invisibilityData$cloak, levels = c(1:2),
                                 labels = c("No Cloak", "Cloak"))

  restructuredData<-reshape2::melt(invisibilityData, id = c("cloak"),
                                   measured = c("mischief1", "mischief2"))
  names(restructuredData)<-c("cloak", "Time", "mischief")
```

In [5]:
```
#------- Normal ANCOVA ------#

  #Normality check
  library(pastecs)
  library(car)
  by(invisibilityData$mischief1, invisibilityData$cloak, stat.desc, basic=F, no

  #Levene's Test
  leveneTest(invisibilityData$mischief2, invisibilityData$cloak, center = media

  #Test whether the IV and covariate are independent
  checkIndependenceModel<-aov(mischief1 ~ cloak, data = invisibilityData)
  summary(checkIndependenceModel)

  #ANCOVA
  invisibilityModel<-aov(mischief2~ mischief1 + cloak, data = invisibilityData)
  Anova(invisibilityModel, type = "III")

  #adjustedMeans<-effects::effect("cloak", invisibilityModel, se = TRUE)
  #summary(adjustedMeans)
  #hoRS<-update(invisibilityModel, .~. + mischief1:cloak)
  #Anova(hoRS, type = "III")
```

In [21]:

```
#------- Robust ANCOVA -------#
ancova(mischief2 ~ cloak + mischief1, data = invisibilityData)
## specifying covariate evaluation points
ancova(mischief2 ~ cloak + mischief1, data = invisibilityData, pts = c(3, 5,
## bootstrap version
ancboot(mischief2 ~ cloak + mischief1, data = invisibilityData,
        nboot=2000, pts = c(3, 5, 7))

#--> Graph plot
library(ggplot2)
boxplot <- ggplot(restructuredData, aes(cloak, mischief))
boxplot + geom_boxplot() + facet_wrap(~Time) +
  labs(x = "Cloak of Invisibility", y = "Number of Mischievous Acts")
```

(5) Robust methods of 2-way ANOVA

Example B:

[Hypothesis]* Album sales can be affected by 3 factors: advert, airplay and attractiveness.** ***(2-tailed)

- Null hypothesis H_0 **:** No prominent relations can be observed among album sales and the 3 factors (all βs = 0).
- Alternative hyp. H_1 **:** The album sales can be predicted by at least one of the 3 factors (βs ≠ 0).

In [22]:
```
#-----< 2-way ANOVA >-----#

  goggles<-read.csv("goggles.csv", header = TRUE)
  t2way(attractiveness ~ gender*alcohol, data = goggles, tr=0.2)
  mcp2atm(attractiveness ~ gender*alcohol, data = goggles, tr = .2) # post hoc
```

```
Call:
t2way(formula = attractiveness ~ gender * alcohol, data = goggles,
    tr = 0.2)

                 value p.value
gender          1.6667   0.209
alcohol        48.2845   0.001
gender:alcohol 26.2572   0.001

Call:
mcp2atm(formula = attractiveness ~ gender * alcohol, data = goggles,
    tr = 0.2)

                    psihat  ci.lower  ci.upper p-value
gender1           10.00000  -6.00223  26.00223 0.20922
alcohol1          39.16667  22.46796  55.86537 0.00001
alcohol2           3.33333 -13.82885  20.49551 0.61070
alcohol3         -35.83333 -52.33911 -19.32755 0.00003
gender1:alcohol1 -25.83333 -42.53204  -9.13463 0.00080
gender1:alcohol2   3.33333 -13.82885  20.49551 0.61070
gender1:alcohol3  29.16667  12.66089  45.67245 0.00025
```

In [24]:
```
#-----< 2-way mixed ANOVA >-----#

  picture<-read.delim("ProfilePicture.dat", header = TRUE)

  #-> data needs to be in long format
  pictureLong <- reshape(picture, direction = "long", varying = list(3:4),
                         idvar = "case", timevar = c("pictype"), times = c("cou
  pictureLong$pictype <- as.factor(pictureLong$pictype)
  colnames(pictureLong)[4] <- "friend_requests"
```

In [25]:

```
#-> Assumption check
by(pictureLong$friend_requests, list(pictureLong$pictype, pictureLong$relatio
leveneTest(pictureLong$friend_requests, interaction(pictureLong$pictype, pict

#-> 2-way mixed-effect subjects ANOVA
FBaov <- aov(friend_requests ~ relationship_status*pictype, data = pictureLon
Anova(FBaov, type='III')
```

```
: alone
: In a Relationship
      median         mean      SE.mean CI.mean.0.95          var      std.dev
   6.0000000    5.6470588    0.3314538    0.7026506    1.8676471    1.3666188
    coef.var     skewness     skew.2SE     kurtosis     kurt.2SE   normtest.W
   0.2420054   -0.2215355   -0.2014884   -1.0466142   -0.4922011    0.9441923
  normtest.p
   0.3713285
------------------------------------------------------------
: couple
: In a Relationship
       median          mean       SE.mean  CI.mean.0.95           var
 3.0000000000  3.2941176471  0.2058823529  0.4364510910  0.7205882353
      std.dev      coef.var      skewness      skew.2SE      kurtosis
 0.8488746876  0.2576941016 -1.1160548604 -1.0150614868  0.6865838772
     kurt.2SE    normtest.W    normtest.p
 0.3228862318  0.7650982586  0.0007005297
------------------------------------------------------------
: alone
: Single
      median         mean      SE.mean CI.mean.0.95          var      std.dev
   8.0000000    7.9130435    0.3971890    0.8237195    3.6284585    1.9048513
    coef.var     skewness     skew.2SE     kurtosis     kurt.2SE   normtest.W
   0.2407230   -0.1456935   -0.1513426   -1.1496031   -0.6149163    0.9285092
  normtest.p
   0.1015505
------------------------------------------------------------
: couple
: Single
      median         mean      SE.mean CI.mean.0.95          var      std.dev
  4.00000000   3.95652174   0.28450262   0.59002233   1.86166008   1.36442665
    coef.var     skewness     skew.2SE     kurtosis     kurt.2SE   normtest.W
  0.34485509   0.48515825   0.50396977  -0.62316034  -0.33332503   0.91604998
  normtest.p
  0.05486059
```

A anova: 2 × 3

	Df	F value	Pr(>F)
	<int>	<dbl>	<dbl>
group	3	2.659338	0.05417303
	76	NA	NA

A anova: 5 × 4

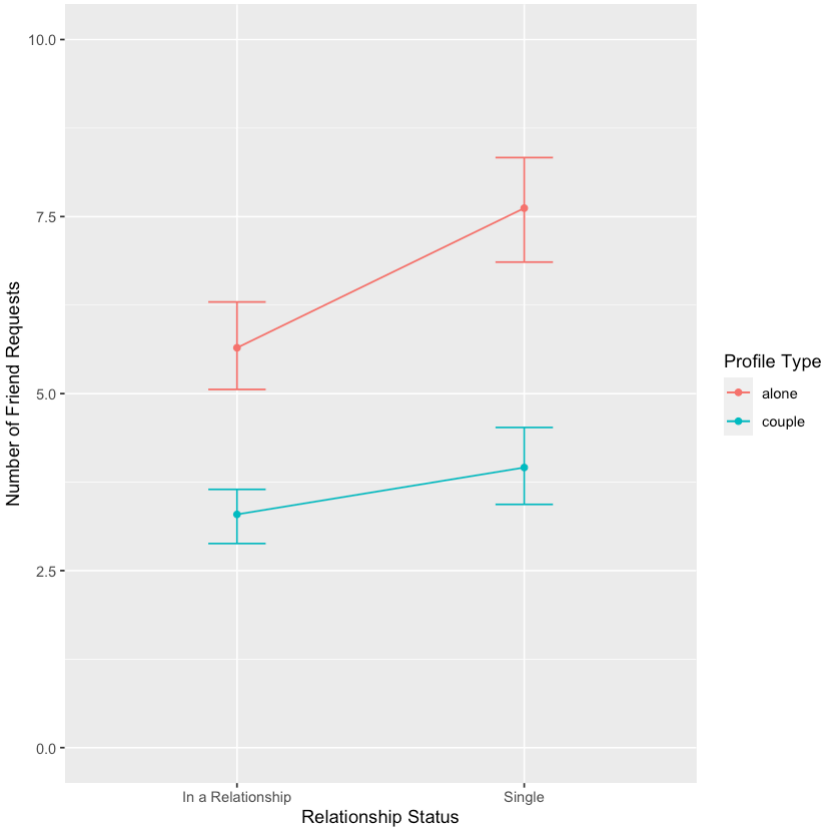
	Sum Sq	Df	F value	Pr(>F)
	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	542.11765	1	254.022013	6.047500e-26
relationship_status	50.19156	1	23.518440	6.419054e-06
pictype	47.05882	1	22.050522	1.154712e-05
relationship_status:pictype	12.56806	1	5.889062	1.760456e-02
Residuals	162.19437	76	NA	NA

In [26]:

```
#-> Robust 2-way mixed-effect subjects ANOVA
#-> "bwtrim" is designed for one between-subjects variable and one within-sub
bwtrim(friend_requests ~ relationship_status*pictype, id = case, data = pictu

## between groups effect only (bootstrap)
sppba(friend_requests ~ relationship_status*pictype, case, pictureLong, nboot
## within groups effect only (bootstrap)
sppbb(friend_requests ~ relationship_status+pictype, case, pictureLong, nboot
## interaction effect only (bootstrap)
sppbi(friend_requests ~ relationship_status*pictype, case, pictureLong, nboot
```

```
Call:
bwtrim(formula = friend_requests ~ relationship_status * pictype,
    id = case, data = pictureLong, tr = 0.2)

                              value df1     df2 p.value
relationship_status         10.7884   1 19.5697  0.0038
pictype                     92.1209   1 19.7954  0.0000
relationship_status:pictype  8.1671   1 19.7954  0.0098

Call:
sppba(formula = friend_requests ~ relationship_status * pictype,
    id = case, data = pictureLong, nboot = 2000)

Test statistics:
                         Estimate
In a Relationship-Single   -1.464

Test whether the corrresponding population parameters are the same:
p-value: 5e-04

Call:
sppbb(formula = friend_requests ~ relationship_status + pictype,
    id = case, data = pictureLong, nboot = 2000)

Test statistics:
             Estimate
alone-couple    3.171

Test whether the corrresponding population parameters are the same:
p-value: 0

Call:
sppbi(formula = friend_requests ~ relationship_status * pictype,
    id = case, data = pictureLong, nboot = 2000)

Test statistics:
                                      Estimate
alone-couple In a Relationship-Single   -1.438

Test whether the corrresponding population parameters are the same:
p-value: 0.02
```

In [27]:

```
#--> Graph plot
profileGraph <- ggplot(pictureLong, aes(relationship_status, friend_requests,
profileGraph + stat_summary(fun = mean, geom = "point") +
  stat_summary(fun = mean, geom = "line", aes(group= pictype)) +
  stat_summary(fun.data = mean_cl_boot, geom = "errorbar", width = 0.2) +
  labs(x = "Relationship Status", y = "Number of Friend Requests", colour = '
  scale_y_continuous(limits = c(0,10))
```

```
Warning message:
“Removed 2 rows containing non-finite values (stat_summary).”
Warning message:
“Removed 2 rows containing non-finite values (stat_summary).”
Warning message:
“Removed 2 rows containing non-finite values (stat_summary).”
```